Lab Assignment 3

Title: Descriptive Statistics - Measures of Central Tendency and variability

PROBLEM STATEMENT:

Descriptive Statistics - Measures of Central Tendency and variability perform the following operations on any open source dataset (e.g., data.csv)

1. Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variables. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable.

2. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset.

THEORY:

What is Statistics?

Statistics is the science of collecting data and analyzing them to infer proportions (sample) that are representative of the population. In other words, statistics is interpreting data in order to make predictions for the population.

There are two branches of Statistics.

- DESCRIPTIVE STATISTICS: Descriptive Statistics is a statistics or a measure that describes the data.
- INFERENTIAL STATISTICS: Using a random sample of data taken from a population to describe and make inferences about the population is called Inferential Statistics.

Descriptive Statistics

Descriptive Statistics is summarizing the data at hand through certain numbers like mean, median etc. so as to make the understanding of the data easier. It does not involve any generalization or inference beyond what is available. This means that the descriptive statistics are just the representation of the data (sample) available and not based on any theory of probability.

Commonly Used Measures

1. **Measures of Central Tendency**
2. **Measures of Dispersion (or Variability)**

Measures of Central Tendency

A Measure of Central Tendency is a one number summary of the data that typically describes the center of the data. These one number summary is of three types.

1. Mean: Mean is defined as the ratio of the sum of all the observations in the data to the total number of observations. This is also known as Average. Thus mean is a number around which the entire data set is spread.

2. Median: Median is the point which divides the entire data into two equal halves. One-half of the data is less than the median, and the other half is greater than the same. Median is calculated by first arranging the data in either ascending or descending order.

- If the number of observations is odd, median is given by the middle observation in the sorted form.
- If the number of observations is even, median is given by the mean of the two middle observations in the sorted form.

An important point to note that the order of the data (ascending or descending) does not affect the median

3. Mode: Mode is the number which has the maximum frequency in the entire data set, or in other words, mode is the number that appears the maximum number of times. A data can have one or more than one mode.

How to calculate summary statistics?

A large number of methods collectively compute descriptive statistics and other related operations on Data Frame. Most of these are aggregations like sum(), mean() etc.

Functions & Description: To calculate Mean, Standard Deviation, Median, Max, and Min we can apply these functions.

Sr.No.	Function	Description
1	count()	Number of non-null observations
2	sum()	Sum of values
3	mean()	Mean of Values
4	median()	Median of Values
5	mode()	Mode of values

6	std()	Standard Deviation of the Values
7	min()	Minimum Value
8	max()	Maximum Value
9	abs()	Absolute Value
10	prod()	Product of Values
11	cumsum()	Cumulative Sum
12	cumprod()	Cumulative Product

Using the 'describe()' Method:-

We can use the describe function to generate the statistics above and apply it to multiple columns simultaneously. It also provides the lower, median and upper percentiles.

Aggregating statistics grouped by category

Using 'groupby()' to Aggregate

Suppose we wanted to know the average runtime for each genre. We can use the 'groupby()' method to calculate these statistics:

The group by method is used to support this type of operations. More general, this fits in the more general split-apply-combine pattern:

- Split the data into groups
- Apply a function to each group independently
- Combine the results into a data structure

The apply and combine steps are typically done together in pandas.

Example:-

titanic.groupby(["Sex", "Pclass"])["Fare"].mean()

Output:

Sex Pclass

female 1 106.125798

2 21.970121

3 16.118810

male 1 67.226127

2 19.741782

3 12.661633

Name: Fare, dtype: float64

Grouping can be done by multiple columns at the same time. Provide the column names as a list to the **groupby()** method.

Count number of records by category-

- The value_counts() method counts the number of records for each category in a column.
- value_counts is a convenient shortcut to count the number of entries in each category of a variable

Procedure-

STEPS:

1. IMPORT REQUIRED LIBRARIES

2. READ CSV FILE (ADULT.CSV) AND (IRIS.CSV)

3. PROVIDE SUMMARY STATISTICS USING PREDEFINED FUNCTION LIKE MEAN(),MEDIAN(),MODE(), DESCRIBE() ETC.

4. CATEGORIZE DATA USING GROUPBY() METHOD AND PROVIDE STATISTICS.

PROGRAM: TO Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variables

```
import pandas as pd

df = pd.read_csv("C:/Users/Admin/Downloads/adult.csv")

print(df)

#summary statistics of age grouped by gender

df.groupby("gender")["age"].describe()

df.groupby("marital-status")["age"].mean()

df.groupby("marital-status")["age"].median()

#grouping can be done on multiple columns

# summary statistics of age grouped by gender & marital-status

df.groupby(["gender","marital-status"])["age"].std()

#summary statistics of age grouped by income

df.groupby("income")["age"].mean()

df.groupby(["income","gender"])["age"].mean()

df.groupby("marital-status")["marital-status"].count()

#Count number of records by category

#The value_counts() method counts the number of records for each category in a column.

df["marital-status"].value_counts()
```

Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset.

Program: (Code without Group by function)

```
import pandas as pd

d = pd.read_csv("C:/Users/Admin/Downloads/Iris.csv")

print('Iris-setosa')

setosa = d['Species'] == 'Iris-setosa'

print(d[setosa].describe())

print('\nIris-versicolor')

setosa = d['Species'] == 'Iris-versicolor'

print(d[setosa].describe())

print('\nIris-virginica')

setosa = d['Species'] == 'Iris-virginica'

print(d[setosa])

print(d[setosa].describe())
```

Program using Group By function:-

```
import pandas as pd

d = pd.read_csv("C:/Users/Admin/Downloads/Iris.csv")

#Species

d.groupby(["Species"])["SepalLengthCm"].mean()

d.groupby(["Species"])["SepalLengthCm"].std()

d.groupby(["Species"])["SepalLengthCm"].describe()

d.groupby(["Species"])["SepalLengthCm"].quantile(q=0.75)

d.groupby(["Species"])["SepalLengthCm"].quantile(q=0.25)

a=d.groupby(["Species"])["SepalLengthCm"].mean()

print(a)

b=d.groupby(["Species"])["SepalLengthCm"].median()

print(b)
```

```
list=[a,b]

print(list)
```

CONCLUSION:

To summarize, here we discussed how to generate summary statistics using the Pandas library. Here, we discussed how to use pandas methods to generate mean, median, max, min and standard deviation. We also saw describe () method which allows us to generate percentiles, in addition to the mean, median, max, min and standard deviation, for any numerical column. Finally, we showed how to generate aggregate statistics for categorical columns.